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RESEARCH

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Calf circumference to body mass index ratio: a new anthropometric indicator for ultrasound based sarcopenic obesity

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Abstract

Objectives Traditional anthropometric measurements, such as calf and arm circumference, may not accurately reflect muscle mass in obese individuals. These limitations highlight the need for more sensitive and practical measures for the early detection of sarcopenic obesity. This study aimed to evaluate the predictive value of the calf circumference to body mass index ratio (CC/BMI) as an anthropometric indicator for ultrasound based sarcopenic obesity, and to compare its effectiveness with calf circumference adjusted for BMI (subtracting 3, 7, or 12 cm from calf circumference measurements corresponding to BMI categories of 25–29.9, 30–39.9, and ≥ 40 kg/m^2, respectively).

Methods A retrospective analysis was conducted on 573 patients aged 65 years and older from a geriatrics outpatient clinic. Exclusion criteria included having advanced dementia, knee or hip replacement, neurodegenerative diseases, decompensated heart failure or the use of muscle-impacting medications. Participants underwent geriatric assessments which included ultrasound measurements of anterior thigh muscle thickness and anthropometric evaluations. Sarcopenia was defined using the STAR index (anterior thigh muscle thickness/BMI ratio) and low handgrip strength. Patients who had both sarcopenia and a BMI ≥ 30 kg/m^2 were categorized as having sarcopenic obesity. CC/BMI, and calf circumference adjusted for BMI were calculated, and their predictive power for sarcopenic obesity was assessed.

Results Of the participants, 37.3% ($n = 214$) were sarcopenic, and 17.3% ($n = 99$) had sarcopenic obesity. Significant positive correlations were observed between the CC/BMI and age, as well as with handgrip strength, STAR index, and walking speed. The CC/BMI demonstrated a significantly higher predictive capability for sarcopenic obesity (AUC: 0.850, cutoff ≤ 1.20, sensitivity: 81.63%, specificity: 75.91%) compared to calf circumference adjusted for BMI (AUC: 0.672, cutoff ≤ 31 cm, sensitivity: 60.20%, specificity: 67.24%).

Conclusion The CC/BMI shows higher sensitivity and predictive value for diagnosing sarcopenic obesity compared to calf circumference adjusted for BMI. It could serve as a practical screening tool for sarcopenic obesity in older adults.

Clinical trial number Not applicable.

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Keywords Sarcopenia, Sarcopenic obesity, Calf circumference, Anthropometric

Introduction

Sarcopenic obesity is a clinical condition characterized by the coexistence of sarcopenia and obesity marked by an excess fat mass [1]. This condition presents a significant clinical challenge, particularly among older adults who face an elevated risk of comorbidities, disability, and mortality [2–4]. Early detection of sarcopenic obesity is essential for implementing timely preventive interventions.

The diagnosis of sarcopenic obesity involves the simultaneous identification of both sarcopenia and obesity. Sarcopenia refers to the loss of muscle mass and strength, while obesity is characterized by excessive fat accumulation. To diagnose sarcopenic obesity, a combination of clinical evaluation, anthropometric measurements, and assessments of muscle strength and muscle mass are utilized. Common tools for measuring body composition and muscle mass include dual-energy X-ray absorptiometry (DEXA) and bioelectrical impedance analysis (BIA). However, alternative methods, such as ultrasound, have also gained recognition due to their portability, ease of use, and ability to assess muscle quality. The anterior thigh, a region particularly vulnerable to sarcopenia, is often targeted in ultrasound measurements as its muscles are among the first to be affected by sarcopenia [5]. By utilizing ultrasound to assess anterior thigh muscle thickness, clinicians can potentially identify sarcopenia at earlier stages, enabling timely intervention and improving patient outcomes.

Recently, the anterior thigh muscle thickness which is calculated as the sum of the rectus femoris and vastus intermedius muscle thickness, has been integrated into the STAR Index. This index, derived by dividing the anterior thigh muscle thickness by body mass index (BMI), defines low muscle mass based on cut-off values of < 1.0 for women and < 1.4 for men, as proposed by Kara et al. [6].

Sarcopenic obesity is an important clinical entity, yet its diagnostic process can be challenging. There is a need for practical, easily measurable clinical tools to predict sarcopenic obesity. In predicting sarcopenia, traditional anthropometric measurements, such as calf circumference (CC), have been widely used, as it is one of the key indicators of muscle mass. However, CC may not accurately reflect muscle mass in the context of sarcopenic obesity, where muscle loss is accompanied by excessive fat accumulation. While these anthropometric indicators are useful in low muscle prediction, their ability to accurately represent muscle mass in sarcopenic obesity is significantly reduced.

Previous research has defined calf circumference values in a diverse population sample along with a BMI-adjustment approach that helps to remove the confounding effects of adiposity and thereby improves CC as a useful clinical estimate of skeletal muscle mass. This calf circumference adjusted for BMI method involved subtracting 3, 7, or 12 cm from calf circumference measurements, corresponding to BMI categories of 25–29.9, 30–39.9, and ≥ 40 kg/m², respectively [7]. While this study has offered valuable insights into the relationship between calf circumference and BMI in predicting sarcopenia, its generalizability to diverse populations remains uncertain, and there are currently no established cut-off values for sarcopenic obesity.

Furthermore, we hypothesize that adjusting calf circumference for BMI, yielding the the calf circumference to body mass index ratio (CC/BMI), may effectively mitigate the confounding effects of excess adiposity, which can distort muscle mass estimates in obese individuals. However, its validity and predictive accuracy in detecting sarcopenic obesity in older adults require further investigation. Given the rising prevalence of both obesity and sarcopenia in aging populations, there is an urgent need for reliable and effective tools to predict sarcopenic obesity. Adjusting calf circumference for BMI offers a promising approach to address this critical clinical challenge.

Given these gaps, there is a pressing need for practical screening methods that can be applied to large older populations. This study aims to evaluate the predictive utility of the CC/BMI as a screening tool for ultrasound based sarcopenic obesity in older adults. Additionally, it seeks to compare the predictive efficacy of CC/BMI with that of calf circumference adjusted for BMI in identifying individuals with sarcopenic obesity.

Methods

A total of 919 patients aged 65 years and older, who attended the geriatrics outpatient clinic between February 2022 and February 2024 and provided informed consent, were evaluated in this study. Exclusion criteria included knee or hip replacement, advanced dementia, cerebrovascular disease, neurodegenerative disease, decompensated heart failure, and use of medications that affect muscle function, such as steroids. Based on these criteria, 84 patients were excluded due to knee or hip replacement, 61 patients due to advanced dementia, and 199 patients due to cerebrovascular disease, neurodegenerative disease, decompensated heart failure, and use of muscle-affecting drugs such as steroids. Consequently, 573 patients were included in the final analysis.

All participants underwent a comprehensive geriatric assessment which included the application of several assessment tools: the Katz Activities of Daily Living (ADL) [8], the Lawton Brody Instrumental Activity of Daily Living (IADL) [9], the Yesavage Geriatric Depression Scale short form (GDS) [10, 11], the Mini Nutritional Assessment Short Form (MNA-SF) [12, 13] and the Clinical Frailty Scale (CFS) [14]. The ADL scale ranges from 0 to 6 points, and the IADL scale from 0 to 8 points, with higher scores indicating greater independence. The GDS ranges from 0 to 15 points with scores of ≥ 5 suggesting the presence of depressive symptoms. The MNA-SF ranges from 0 to 14 points with scores below 12 indicating a risk of malnutrition. The CFS ranges from 1 to 9 points, with 1 indicating robustness and 9 denoting terminal frailty.

The presence of comorbidities was confirmed through review of hospital medical records and current pharmacological treatments. Hypertension was defined as a previously established diagnosis or current use of antihypertensive medications. Diabetes mellitus was identified based on a known diagnosis, fasting plasma glucose ≥ 126 mg/dL, HbA1c ≥ 6.5%, or the use of antidiabetic medications. Coronary artery disease was defined by a history of myocardial infarction, prior coronary revascularization, or a diagnosis confirmed by a cardiologist. Chronic obstructive pulmonary disease was identified based on clinical history of respiratory disease and/or prior pulmonary function tests demonstrating an obstructive pattern (FEV1/FVC < 0.7), along with the use of inhaled bronchodilators.

Anthropometric measurements and ultrasound evaluations of anterior thigh muscle thickness were performed on all participants. A portable height meter was employed for the purpose of measuring height. The patient was asked to remove any accessories, such as hats and shoes, and to stand upright during the measurement. The measurement was taken from the vertex. For the purpose of measuring weight, the Tanita device was adjusted to subtract 1 kg for clothing weight. The patient was asked to remove their shoes before the measurement was taken. The same Tanita device was used for all participants who were weighed in a fasting state. Waist circumference was measured at the level of the umbilicus while the patient was standing and without clothing. Hip circumference was measured with a non-elastic tape at the widest part of the buttocks, also in a standing position. Arm circumference was measured at the midpoint between the acromion and olecranon processes, with the arm relaxed and hanging freely at the side.

Muscle strength was assessed using a handgrip dynamometer (Takei), with three measurements taken from the dominant hand, and the highest value recorded. Participants were classified as having low muscle strength if their measurements fell below the threshold (< 32 kg for men and < 22 kg for women) defined for the Turkish population [15].

The diagnosis of sarcopenia was determined by a low handgrip strength in addition to a low STAR (Sonographic Thigh Adjustment Ratio) index. The term sarcopenic obesity was defined as patients presenting with a BMI above 30 kg/m² and concomitant sarcopenia. The STAR index is calculated by dividing the anterior thigh muscle thickness (measured using ultrasonography by a geriatrician certified in musculoskeletal ultrasonography) by BMI, with cutoff values of < 1.4 for men and < 1.0 for women [6]. For the measurement of the thickness of the anterior thigh muscle, the participant lay supine in a relaxed position with the hip in neutral and the knee in extension. The measurements were taken from the right leg. The midpoint of the distance measured from the antero-superior iliac spine to the superior pole of the patella was marked. The measurement was taken from this marked area. The probe was placed on the skin with minimal pressure applied to ensure a clear image of the rectus femoris, vastus intermedius, and the femur. The anterior thigh muscle thickness was measured as the total thickness of the rectus femoris and vastus intermedius muscles in millimeters (mm) (Fig. 1). Physical performance was assessed by measuring walking speed over a six meter distance. Walking speeds < 0.8 m/second were classified as slow [16].

All anthropometric and ultrasound-based measurements were independently performed by four geriatricians with prior training and experience in musculoskeletal ultrasound. These clinicians regularly conduct comprehensive geriatric assessments, including body composition evaluations, in daily practice. To assess the consistency among raters, we calculated inter-rater reliability using intraclass correlation coefficients (ICCs) based on a random subsample of 30 participants. The ICC values for key measurements — including calf circumference and anterior thigh muscle thickness — ranged from 0.93 to 0.97, indicating excellent agreement among raters.

CC/BMI, calculated as calf circumference in centimeters divided by BMI (kg/m²). "Calf circumference adjusted for BMI was calculated by subtracting 3, 7, or 12 cm for BMI categories of 25–29.9, 30–39.9, and ≥ 40 kg/m², respectively [7]. The predictive capacity of the CC/BMI for sarcopenic obesity was analyzed and compared with that of calf circumference adjusted for BMI.

Statistical analysis

The research data was analyzed using the SPSS 22.0 statistical software package. Categorical variables were presented as numbers and percentages, while continuous variables were displayed as means ± standard deviations

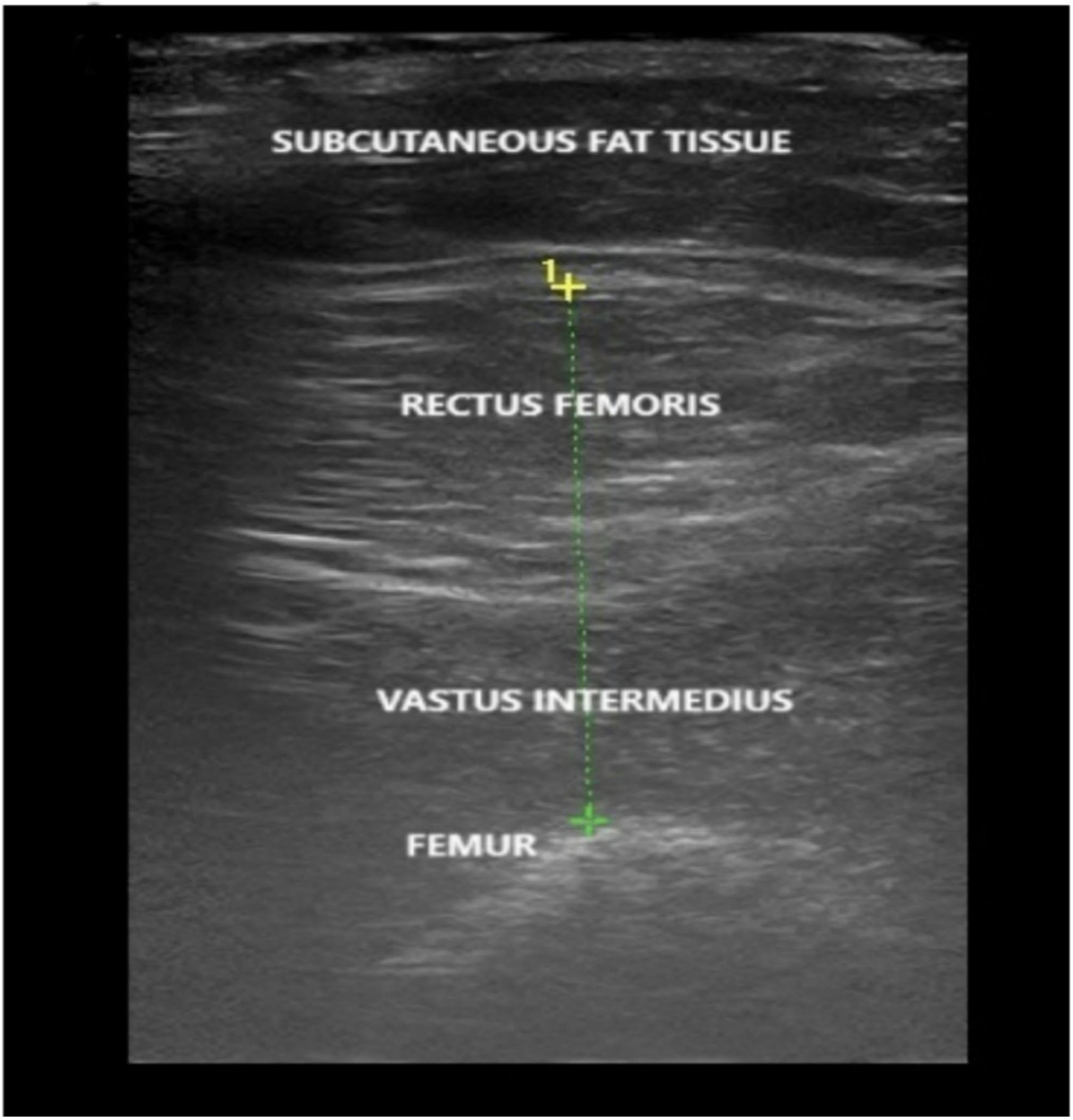

Fig. 1 Ultrasonographic Measurement of Anterior Thigh Muscle Thickness

for normally distributed data and as medians (min, max) for non-normally distributed data. Chi-square tests were used to compare the frequency of categorical variables between groups.

The normality of continuous variables was assessed using visual and analytical methods. Visual assessments included histograms and probability plots, while analytical methods comprised the Kolmogorov-Smirnov and Shapiro-Wilk tests. For normally distributed continuous variables, independent-samples T-test were used to compare groups, while the Mann-Whitney U test was used for non-normally distributed variables. The predictive ability of the CC/BMI and calf circumference adjusted for BMI for sarcopenic obesity was evaluated using Receiver Operating Characteristic (ROC) curve analysis, with MedCalc Statistical Software v15.2 (MedCalc Software, Ostend, Belgium). The optimal thresholds were selected based on the Youden Index to maximize both sensitivity and specificity for identifying sarcopenic obesity. In this study, the statistical significance level was accepted as $p < 0.05$.

Results

The mean age of the patients was 74.08 ± 5.94 years. Out of the total, 62.3% ($n = 357$) were female. 39.6% of the patients were high school and university graduates. Sarcopenia was identified in 37.4% ($n = 214$) of the patients, and 17.3% ($n = 99$) met the criteria for sarcopenic obesity. Individuals with sarcopenic obesity had significantly higher waist, hip, weight, arm, and calf circumferences compared to those without ($p < 0.001$). They also exhibited lower levels of calf circumference adjusted for BMI, CC/BMI, height, handgrip strength, STAR index ($p < 0.001$), and walking speed (p: 0.002). Hypertension and coronary artery disease were significantly more prevalent in those with sarcopenic obesity ($p < 0.001$, p: 0.003 respectively) (Table 1).

Significant positive correlations were observed between the CC/BMI and age, as well as with handgrip strength and walking speed (Table 2).

The CC/BMI demonstrated predictive capability in identifying sarcopenic obesity with an AUC of 0.850, a cutoff value of ≤ 1.20, a positive predictive value of 41.70%, a negative predictive value of 95.1%, a sensitivity of 81.63%, and a specificity of 75.91% (p: <0.0001). However, calf circumference adjusted for BMI had an AUC of 0.672, a cutoff value of ≤ 31 cm, a positive predictive value of 28%, a negative predictive value of 88.9%, a sensitivity of 60.2%, and a specificity of 67.24% for predicting sarcopenic obesity (p: <0.0001) (Table 3). When comparing the areas under the curves, the CC/BMI showed significantly higher sensitivity than calf circumference adjusted for BMI in diagnosing sarcopenic obesity (p: <0.0001) (Figs. 2, 3 and 4).

Table 1 General characteristics of the participants according to sarcopenic obesity

	Total (*n*:573)	**With Sarcopenic obesity (*n*:99)**	**Without Sarcopenic obesity (*n*:474)**	***p***
Sex, n (%)				
Female	357 (62.3)	80 (22.40)	277 (77.60)	< 0.001
Male	216 (37.7)	19 (8.80)	197 (91.20)	
Age, mean (SD)	74.08 (5,94)	73.58 (6.12)	74.18 (5.91)	0.475
Height, cm, mean (SD)	157.39 (9.25)	152.02 (6.90)	158.49 (9.29)	< 0.001
Weight, kg, mean (SD)	70.83 (13.35)	78.67 (10.62)	69.22 (13.29)	< 0.001
BMI, kg/m^2, median (min-max)	28 (13.90–46.30)	33.45 (30-44.30)	27.10 (13.90–46.30)	< 0.001
Waist circumference, cm, mean (SD)	99.01 (12.46)	107.96 (10.17)	97.17 (12.09)	< 0.001
Hip circumference, cm, mean (SD)	106.96 (10.18)	115.50 (9.87)	105.21 (9.33)	< 0.001
Arm circumference, cm, mean (SD)	28.96 (3.72)	31.80 (3.46)	28.37 (3.50)	< 0.001
Calf circumference, cm, mean (SD)	36.07 (3.88)	38.16 (2.98)	35.64 (3.91)	< 0.001
Calf circumference adjusted for BMI, cm, mean (SD)	32.48 (3.29)	30.93 (2.82)	32.80 (3.30)	< 0.001
CC/BMI, median (min-max)	1.27 (0.73–2.27)	1.12 (0.84–1.37)	1.31 (0.73–2.27)	< 0.001
Handgrip strength, kg, median (min-max)	20.35 (7.10–48)	17.30 (8.10–31)	21.50 (7.10–48)	< 0.001
Anterior Thigh Muscle Thickness, mm, mean (SD)	32.87 (7.22)	30.31 (5.51)	33.40 (7.42)	< 0.001
Usg star index, mean (SD)	1.16 (0.26)	0.88 (0.15)	1.21 (0,25)	< 0.001
Gait Speed, m/s, mean (SD)	1.02 (0,29)	0.94 (0.31)	1.03 (0.29)	0.002
ADL, median (min-max)	6 (4–6)	6 (4–6)	6 (4–6)	0.471
IADL, median (min-max)	8 (1–8)	8 (2–8)	8 (1–8)	0.066
MNA, median (min-max)	13 (3–14)	13 (7–14)	13 (3–14)	0.925
GDS, median (min-max)	1 (0–14)	1 (0–10)	1 (0–14)	0.856
Number of drugs, median (min-max)	3 (0–11)	3.5 (0–10)	3 (0–11)	0.017
Diabetes Mellitus, n (%)	186 (32.5)	41 (22)	145 (78)	0.036
Hypertension, n (%)	400 (69.8)	85 (21.30)	315 (78.80)	< 0.001
COPD, n (%)	20 (3.7)	2 (10)	18 (90)	0.551
CAD, n (%)	121 (21.1)	32 (26.40)	89 (73.60)	0.003

BMI: Body Mass Index, CC: Calf Circumference ADL: Activities of Daily Living, IADL: Instrumental Activities of Daily Living, MNA: The Mini Nutritional Assessment, GDS: Geriatric Depression Scale, COPD: Chronic Obstructive Pulmonary Disease, CAD: Coronary Artery Disease

Table 2 Correlations of calf circumference BMI ratio and calf circumference adjusted for BMI

	Calf/bmi		CC adjusted for BMI	
	Rho	***p***	**Rho**	***p***
Age	0.077	0.067	-0.118	0.005
Handgrip strength	0.241	< 0.001	0.359	< 0.001
Anterior Thigh Muscle Thickness	-0.137	0.001	0.160	< 0.001
Gait speed test	0.117	0.005	0.206	< 0.001

Rho: Pearson's Correlations Coefficient

Table 3 Comparison of ROC curves

	Cut-off	AUC	SE	*p*	95% CI	Sensitivity	Specificity	+PV	-PV
CC/BMI	≤ 1.20	0.850	0.0180	< 0,0001	0,818-0,879	81.63	75.91	41.70	95.1
Calf circumference adjusted for BMI	≤ 31	0.672	0.0294	< 0,0001	0,631-0,710	60.2	67.24	28	88.9

ROC: Receiver Operating Characteristic, AUC: Area Under The Curve, SE: Standart Error, PV: Positive Predictive

An additional analysis was performed using the handgrip strength cut-off values recommended by the European Working Group on Sarcopenia in Older People (EWGSOP2). In this analysis, the CC/BMI ratio retained its strong predictive performance for identifying sarcopenic obesity. At a cut-off value of ≤ 1.21, the CC/BMI yielded an area under the curve (AUC) of 0.824, with a high sensitivity of 86.27% and a negative predictive value of 98%, indicating excellent ability to rule out sarcopenic obesity. The specificity was 67.25%, and the positive predictive value was 20.8% ($p < 0.0001$). These findings confirm the clinical utility of the CC/BMI ratio as a reliable and sensitive screening tool for sarcopenic obesity, even when international criteria for muscle strength are applied.

Discussion

The findings of this study demonstrate that the CC/BMI exhibits superior sensitivity and positive predictive value compared to calf circumference adjusted for BMI, confirming its efficacy in identifying sarcopenic obesity. Specifically, the study yielded an AUC of 0.850, with a sensitivity of 81.63% and a negative predictive value of 95.1% for a CC/BMI ≤ 1.20. Early detection of sarcopenic obesity is essential, as it enables timely interventions to reduce the associated risks of disability and mortality.

Furthermore, the significant correlations between the CC/BMI and key clinical parameters such as handgrip strength and gait speed underscore its broader clinical relevance. These findings suggest that the CC/BMI is not only a robust marker of sarcopenic obesity but also a valuable indicator of musculoskeletal health and functional capacity, both of which are crucial determinants of quality of life in older adults.

By contrast, the method of adjusting calf circumference for BMI demonstrated weaker predictive performance, with an AUC of 0.672 and a sensitivity of 60.2%. This suggests that BMI-based adjustments may inadequately account for variations in muscle mass, adiposity, and body composition, particularly in the elderly. The superior discriminative ability of the CC/BMI is likely attributable to its incorporation of both fat and muscle parameters.

Recent studies have shown that sarcopenic obesity is associated with hyperglycaemia, hypertension, dyslipidemia, insulin resistance and an elevated cardiovascular risk profile [3, 17]. In our study, the prevalence of hypertension and cardiovascular disease was notably higher in the sarcopenic obese group, consisting with the existing literature [2].

There are several recommended screening methods for sarcopenic obesity. According to the Consensus Statement by the European Society for Clinical Nutrition and Metabolism (ESPEN) and the European Association for the Study of Obesity (EASO), screening should include surrogate parameters for sarcopenia (such as clinical symptoms, clinical suspicion, or questionnaires like SARC-F in older adults), alongside increased BMI or waist circumference. These measurements should be determined using ethnicity-specific thresholds, and patients meeting both criteria should proceed to the diagnostic stage [1].

Despite the availability of multiple screening methods for sarcopenic obesity, each has notable limitations. Varying cut-off points for BMI and waist circumference have been proposed, with differences influenced by ethnicity, the presence of comorbid conditions such as hypertension and dyslipidemia, and mortality risk assessments. Establishing standardized reference ranges for different racial or ethnic groups remains a challenge. For instance, the World Health Organization (WHO) defines obesity as a BMI ≥ 30 kg/m^2 due to its association with increased mortality risk [18]. However, BMI alone may not accurately reflect obesity prevalence, as studies have shown it often underestimates body fat when compared to direct measurement methods, such as dual-energy X-ray absorptiometry (DEXA). Furthermore, its diagnostic accuracy decreases with advancing age in both men

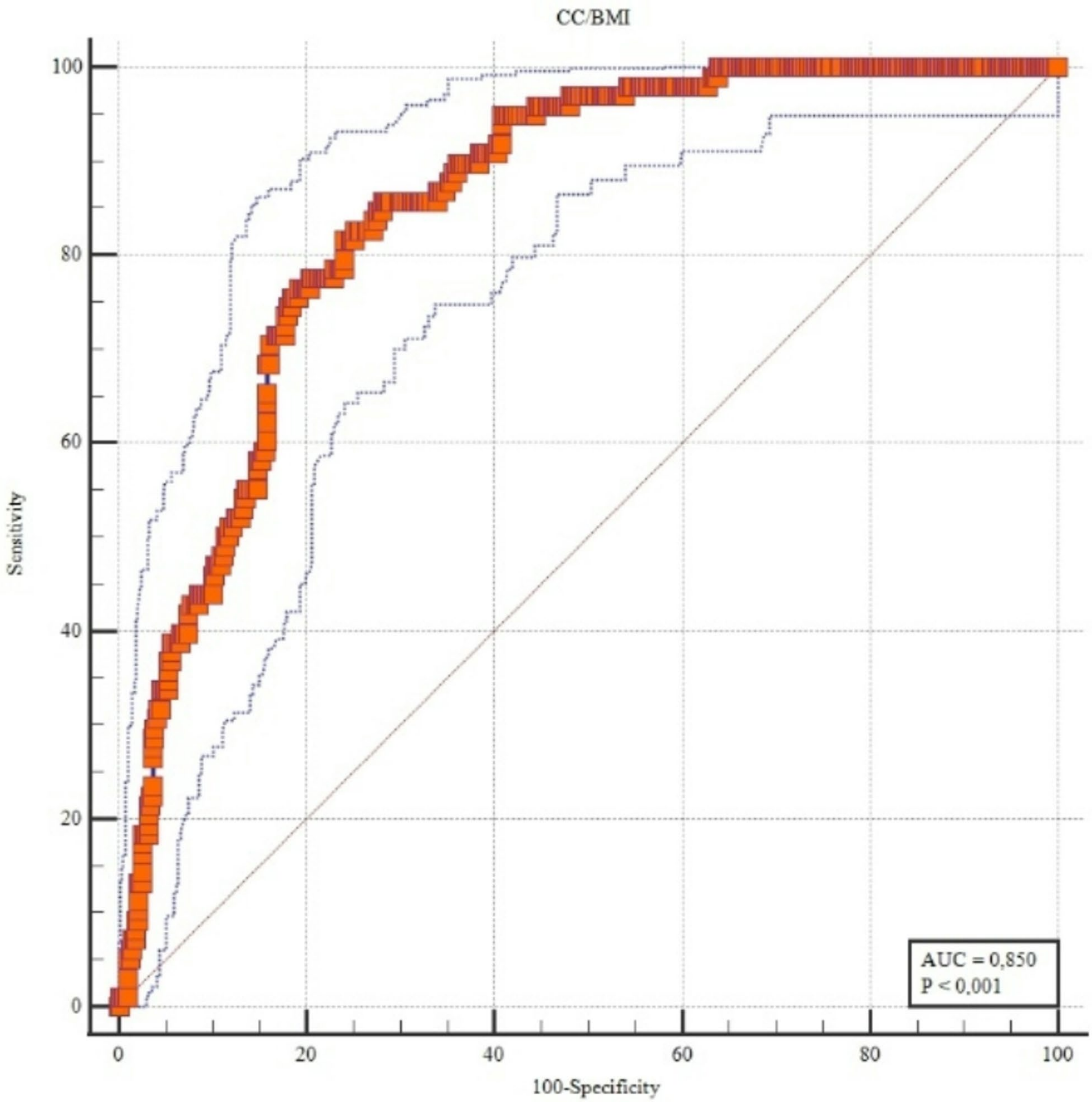

Fig. 2 CC/BMI Roc Curve

and women, evidenced by declining concordance indices [19, 20]. It is important to note that body composition can change significantly even when BMI remains within a certain range. For example, an individual may have a BMI and/or waist circumference within the normal range, but have increased adipose tissue and decreased muscle mass, resulting in sarcopenic obesity [20, 21].

Screening for sarcopenic obesity requires clinical suspicion based on factors such as advanced age, chronic inflammatory or endocrine diseases, recent hospitalization, surgery, immobility, and rapid weight changes [1]. Clinical signs and symptoms can be misleading in sarcopenia screening because they are often nonspecific. Although questionnaires like the SARC-F are commonly used to screening sarcopenia, their sensitivity is often inadequate, as demonstrated by a sensitivity of only 29.5% [22]. The modified SARC-Calf questionnaire improves sensitivity by incorporating calf circumference, yet its role in detecting sarcopenic obesity remains unclear [1, 23].

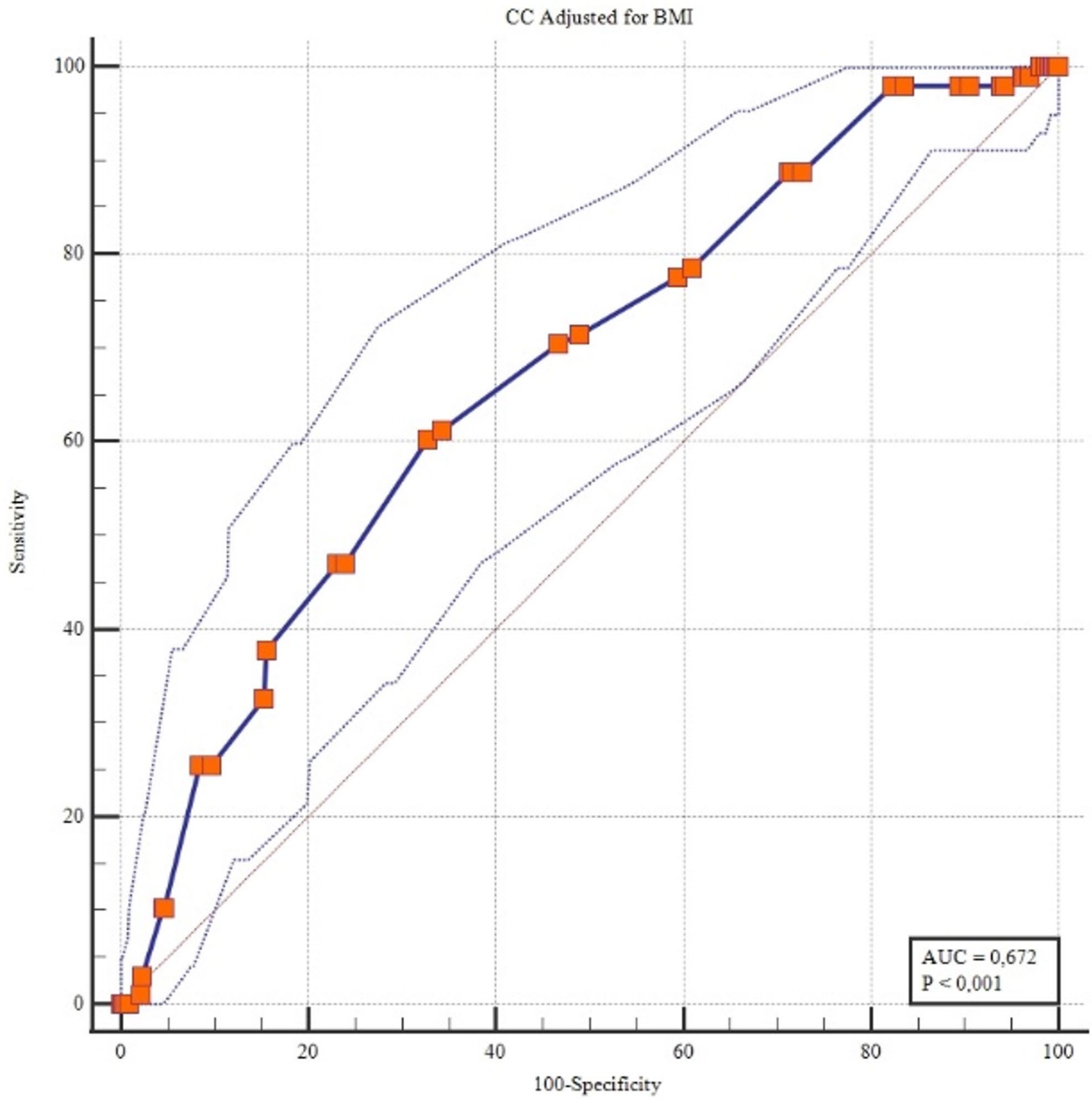

Fig. 3 Calf circumference adjusted for BMI Roc Curve

Calf circumference is another anthropometric measurement for sarcopenia screening, with population-specific cut-offs has already been established also in our country. However, the cut-off value for predicting sarcopenia in obese individuals remains unknown. Recently, the utility of calf circumference in patients with obesity has garnered significant attention. A recent study proposed using adjusted calf circumference, which is calculated by adding 4 cm for individuals with a BMI of less than 18.5 or subtracting 3 cm, 7 cm, or 12 cm for those with a BMI of 25–29, 30–39, and 40 or above, respectively, from the initial calf circumference measurement [7]. This adjustment suggests that calf circumference, when modified for BMI, may correlate better with low muscle mass as determined by DEXA. However, the sensitivity of BMI-adjusted calf circumference in predicting sarcopenic obesity has not yet been tested.

Recognizing the limitations of current screening methods is crucial, as these approaches impact clinical decision-making [23].

The findings of our study suggest that the CC/BMI is an effective method for screening sarcopenic obesity,

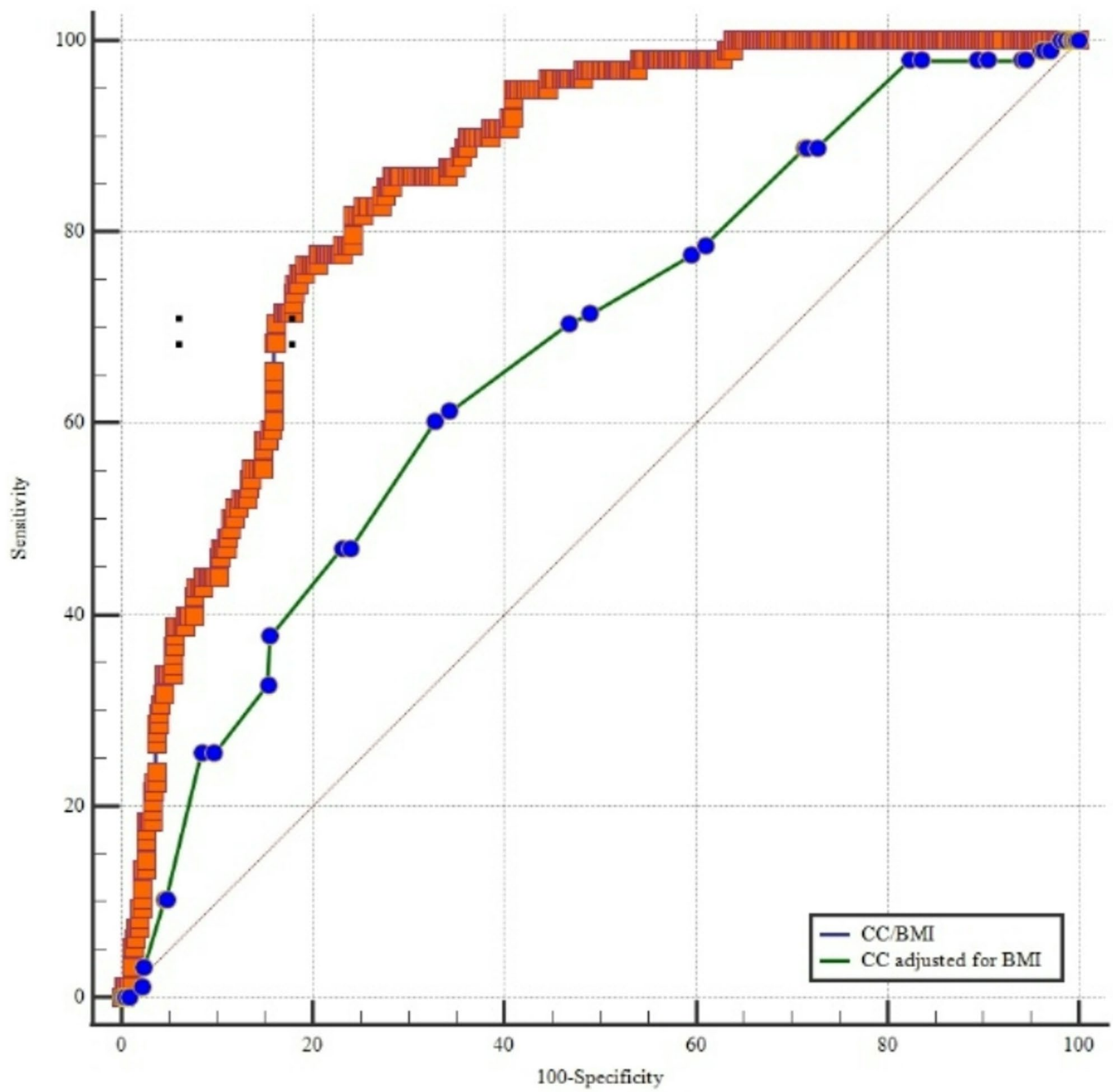

Fig. 4 Comparison Roc Curve

demonstrating notable efficacy compared to calf circumference adjusted for BMI. This method can be implemented without the need for specialized equipment or extensive training, making it easily integrable into the standard geriatric assessment process. Utilizing the CC/BMI in clinical practice may facilitate early intervention and ultimately enhance the health and functional status of older adults at risk of sarcopenic obesity.

There are several notable strengths of this study. To the best of our knowledge, this is the first study to introduce the CC/BMI as a practical and innovative anthropometric tool for detecting sarcopenic obesity and offers a simple, non-invasive, and cost-effective screening option. The use of ultrasound-derived anterior thigh muscle thickness and the STAR index add objective and reliable measures of muscle mass, enhancing the accuracy of sarcopenia diagnosis. The inclusion of multiple parameters, such as handgrip strength and walking speed, and the scales of daily living activities allows for a more holistic evaluation of sarcopenic obesity, further strengthening the study's methodology and relevance to clinical practice. Additionally, by comparing the CC/BMI to calf circumference adjusted for BMI, the study provides valuable insight into the relative effectiveness of different methods for detecting sarcopenic obesity, showing that CC/BMI has superior predictive capability.

However, the study also has limitations. The study was conducted in a geriatrics outpatient clinic, which may limit the generalizability of the findings to other populations or settings, such as community-dwelling older adults or those in institutionalized care. Exclusion of individuals with advanced dementia, neurodegenerative diseases, and other conditions may result in a healthier study cohort, limiting the applicability of findings to more frail or diverse populations. To gain a more comprehensive understanding, future research should include diverse populations from multiple centres and regions. In addition, an increase in BMI does not always mean an increase in fat tissue. The inability to assess body fat is one of the limitations of our study.

Another limitation of our study is the use of grip strength cut-off values specific to the Turkish population. While these values were developed and validated based on a large cohort from Türkiye and are relevant for internal consistency and clinical applicability within this context, they may limit the external validity of our findings. The use of population-specific thresholds may reduce the generalizability of our results to other ethnic or regional groups. To address this, we performed an additional analysis using the European Working Group on Sarcopenia in Older People (EWGSOP2) cut-off values, which yielded similar predictive accuracy for sarcopenic obesity, further supporting the robustness of our findings. Nevertheless, further studies using internationally accepted cut-off values across diverse populations are necessary to validate the applicability of our results in broader contexts.

One important limitation of our study is the lack of direct measurements of fat mass and fat-free mass (FFM), which are considered more accurate indicators for assessing body composition and diagnosing obesity. Although we observed the coexistence of low muscle mass and obesity—potentially suggesting an increased fat mass relative to FFM—this could not be confirmed in the absence of objective body composition analyses. Therefore, the lack of fat mass and FFM data may have limited the accuracy of our findings related to sarcopenic obesity. Future studies incorporating direct measurements of body composition are warranted to validate and expand upon our results.

Conclusion

Sarcopenic obesity, which is associated with an increased risk of morbidity and mortality, should be regarded as a significant public health concern [2]. It is of significant importance to diagnose this condition at an early stage. A review of the literature reveals no studies investigating the CC/BMI as an anthropometric marker. This study introduces the CC/BMI as a practical method in sarcopenic obesity into the literature, marking a first in this field of research.

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Author contributions
N. Karakurt, H. D. Varan equally contributed to the conception and design of the research; F. Gungor, A. Fadıloglu, E. Çataltepe, E. Çeker, Z. Ulger contributed to the acquisition and analysis of the data; N. Karakurt, H. D. Varan contributed to the interpretation of the data; and: N. Karakurt, and H. D. Varan drafted the manuscript. All authors critically revised the manuscript, agree to be fully accountable for ensuring the integrity and accuracy of the work, and read and approved the final manuscript.

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Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
The Human Ethics and Consent to Participate statements are available for consultation. Approval for the ethical conduct of the research project was granted by the Gazi University Clinical Research Ethics Committee on 27 May 2024. The document application number is ca1a3223-e0b9-42f7-820b-b53e6b9f1a43. The research was conducted in accordance with the principles of the Declaration of Helsinki, and informed consent was obtained from all participants.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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